

The Visual Guide to Matrix Decompositions

Take one complicated matrix apart into a few simple, honest pieces — and finally *see* what eigenvalues, eigenvectors and singular values are really doing. We climb a ladder: from the cleanest case (spectral) to the most general (SVD).

Flow: Decomposition $\rightarrow$ Spectral $\rightarrow$ Eigen $\rightarrow$ SVD $\rightarrow$ Low-Rank • **Style:** intuition first, full rigor second • every formula fully solved

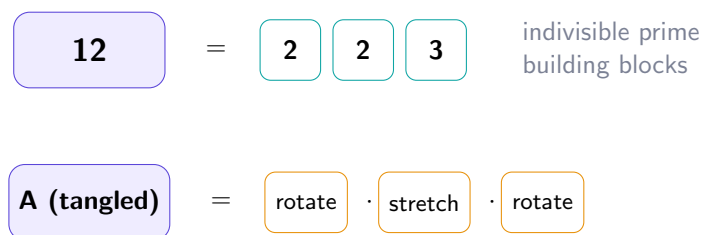
Contents: 1. Decomposition explained easily • 2. Spectral decomposition • 3. Eigen decomposition • 4. SVD • 5. Low-rank approximation & spectral norm • 6. The unifying picture

Part 0 | Decomposition, Explained Easily

THE BIG PICTURE — WHY TAKE A PERFECTLY GOOD MATRIX APART

You already know how to decompose things. The number **12** is hard to reason about until you write it as $2 \times 2 \times 3$. Suddenly everything is obvious: it is even, divisible by 3, its square root is $2\sqrt{3}$. You did not change the number — you revealed what it was *made of*.

Matrix decomposition is exactly this idea, applied to transformations. A matrix is not just a grid of numbers — it is a *verb*. It takes vectors and rotates, stretches, squishes, or shears them. A raw matrix mixes all those actions into one tangled bundle. Decomposition factors that bundle into a short sequence of *pure, simple moves* you can actually picture.



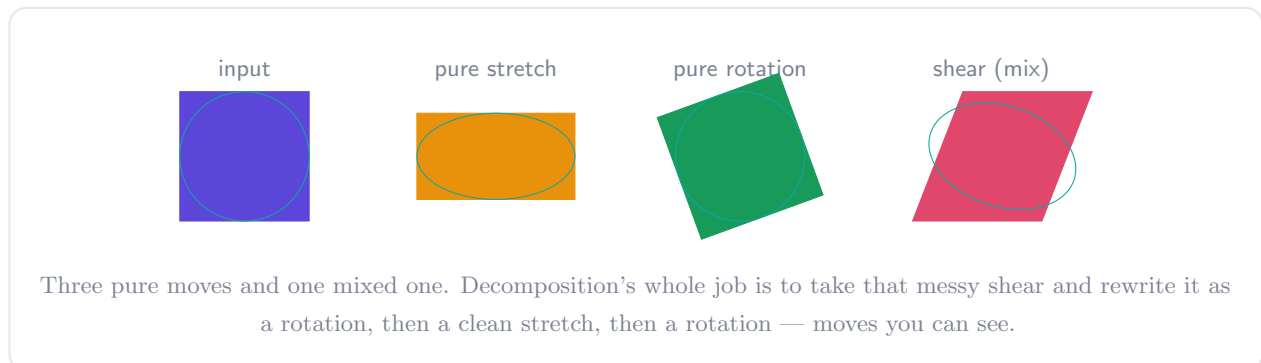
Same trick, two worlds. Factoring a number exposes its primes; factoring a matrix exposes the handful of pure rotations and stretches hiding inside it.

KEY IDEA

The one sentence to remember. A matrix decomposition rewrites A as a product of matrices that each do *one easy thing* — usually **rotate**, **scale along axes**, and **rotate back**. The product still equals A ; it is just written in a language you can read.

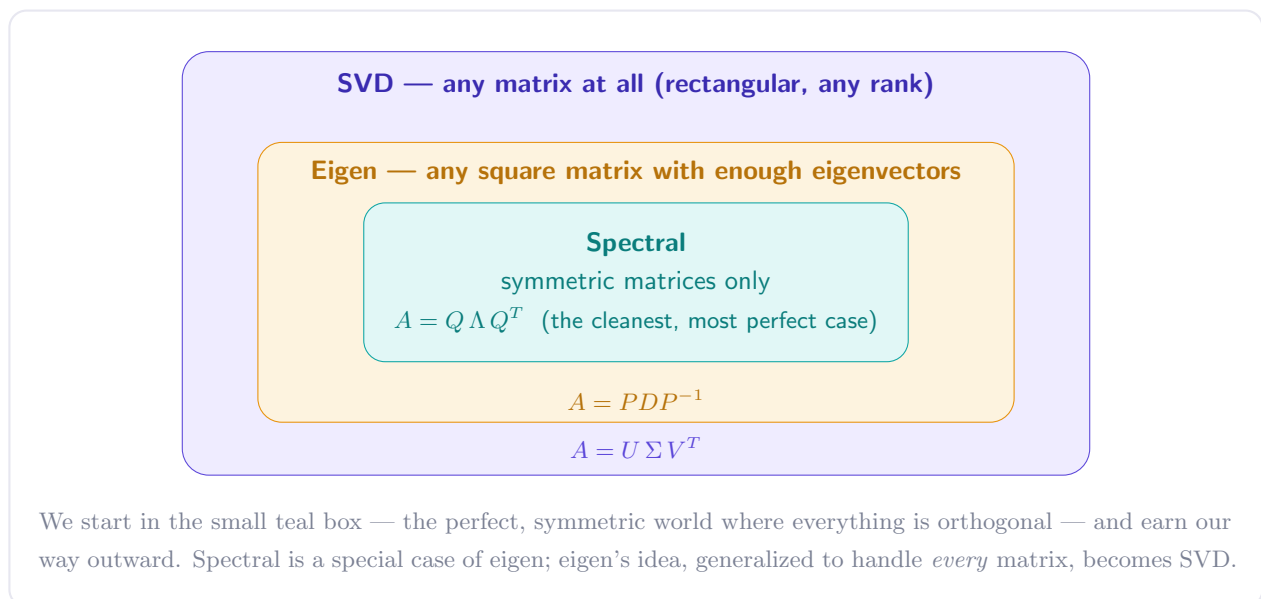
A matrix is a transformation — watch it move

Feed a matrix the same little square grid and a unit circle, and look at what comes out. That output *is* the matrix’s personality.



The ladder of generality — our roadmap

Here is the punchline of the entire lecture in one image. The three decompositions are not three unrelated tricks; they are **the same idea applied to progressively harder matrices**. Each rung relaxes a restriction the previous one needed.



Why climb in this order? Because each step teaches one new lesson:

1. **Spectral** shows the dream scenario: perpendicular axes, real scalings, no distortion — a matrix that is pure stretch in disguise.
2. **Eigen** asks: what if the natural axes *are not* perpendicular? We keep the “scale along special directions” idea but pay a price (the inverse P^{-1} instead of a clean transpose).
3. **SVD** asks the boldest question: what if the matrix is not even square, so “eigenvectors” do not exist? The fix is elegant — use *two* sets of perpendicular axes, one for input and one for output. That single idea works for every matrix in existence.

WHY IT MATTERS

This is not abstract bookkeeping. Spectral decomposition is the engine of **PCA** and quadratic optimization; SVD powers **image & data compression, recommender systems, latent semantic analysis, noise removal**, and the **pseudo-inverse** behind least-squares regression. Master the geometry here and those applications stop being magic.

Part 1 | Spectral Decomposition

THE PERFECT CASE — SYMMETRIC MATRICES, FLAWLESS GEOMETRY, EVERYTHING PERPENDICULAR

We begin at the top of the food chain: the matrices that behave *perfectly*. A **symmetric matrix** (one where $A = A^T$) is secretly nothing but a pure stretch along a set of **perpendicular** axes. No twist, no shear, no surprises. Understand this case and the rest of the lecture is just “what breaks when we lose this perfection?”

INTUITION

The trampoline. Picture a circular trampoline. Grab it at two perpendicular points and stretch: one axis grows by a factor λ_1 , the perpendicular axis by λ_2 . The circle becomes an ellipse whose axes *line up perfectly* with the directions you pulled. That is exactly what a symmetric matrix does. The pull-directions are its **eigenvectors** (they stay perpendicular and never rotate); the stretch-factors are its **eigenvalues** λ_i .

The defining equation: a direction the matrix cannot rotate

An **eigenvector** q is a stubborn direction: when the matrix acts on it, it is not knocked off its line — it only gets longer or shorter (or flipped):

$$Aq = \lambda q \iff \text{“}A \text{ acts on } q \text{ as plain multiplication by the number } \lambda\text{.”}$$

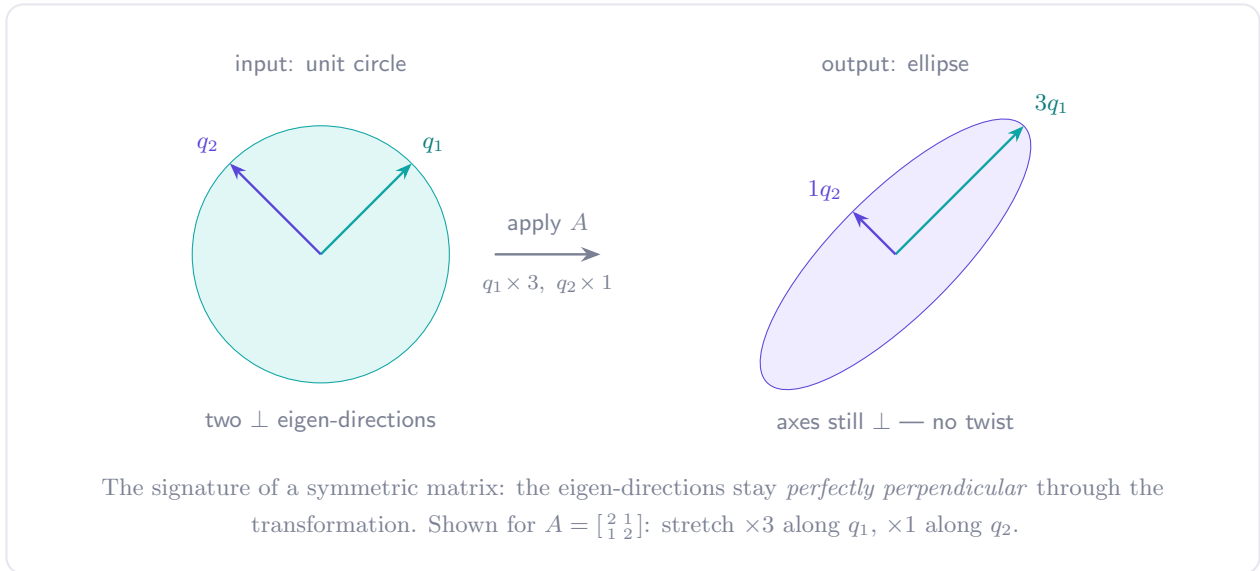
Every other vector gets rotated *and* stretched in a tangled way. Eigenvectors are the rare directions where the tangle disappears. They are the natural axes of the transformation.

Stacking the eigenvectors: $A = Q\Lambda Q^T$

For a symmetric matrix the **Spectral Theorem** guarantees a full set of eigenvectors that are **orthonormal** (perpendicular, unit length). Stack them as columns of Q and put the eigenvalues on the diagonal of Λ :

$$A = Q\Lambda Q^T \quad Q^T Q = I.$$

Read right-to-left: Q^T **rotates** into the eigen-axes, Λ **scales** each axis by its eigenvalue, Q **rotates back**. Because Q is orthogonal, “undo the rotation” is just the transpose Q^T — no inverse needed. *This clean transpose is the luxury only symmetric matrices enjoy.*



The most beautiful reading: a sum of projections

Multiply QAQ^T out and it unpacks into a **weighted sum of simple rank-1 pieces**:

$$A = QAQ^T = \lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T + \cdots + \lambda_n q_n q_n^T.$$

Each block $q_i q_i^T$ is a **projector** — “flatten everything onto the line through q_i .” So a symmetric matrix is literally “*project onto axis 1 and amplify by λ_1 , plus project onto axis 2 and amplify by λ_2 , ...*” The eigenvalues are an honest **importance ranking**. (Hold onto this — it is the seed of SVD and compression in Part 4.)

Problem.

Find $A = QAQ^T$ for $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. It is symmetric, so the Spectral Theorem promises real eigenvalues and orthonormal eigenvectors.

Step 1 — Eigenvalues from $\det(A - \lambda I) = 0$.

Expanding along the first row and factoring the shared $(2 - \lambda)$:

$$(2 - \lambda)[(2 - \lambda)^2 - 2] = 0 \implies (2 - \lambda)(\lambda^2 - 4\lambda + 2) = 0.$$

The first factor gives $\lambda = 2$; the quadratic gives $\lambda = \frac{4 \pm \sqrt{8}}{2} = 2 \pm \sqrt{2}$. Ordered:

$$\lambda_1 = 2 + \sqrt{2} \approx 3.414, \quad \lambda_2 = 2, \quad \lambda_3 = 2 - \sqrt{2} \approx 0.586.$$

All real and positive — the hallmark of a symmetric (positive-definite) matrix.

Step 2 — Eigenvectors (a basis for each null space).

Solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue, then normalize to unit length:

$$\lambda_2 = 2: \quad -y = 0, \quad x = -z \implies q_2 = \frac{1}{\sqrt{2}}(1, 0, -1)^T,$$

$$\lambda_1 = 2 + \sqrt{2}: \quad y = -\sqrt{2}x, \quad x = z \implies q_1 = \frac{1}{2}(1, -\sqrt{2}, 1)^T,$$

$$\lambda_3 = 2 - \sqrt{2}: \quad y = \sqrt{2}x, \quad x = z \implies q_3 = \frac{1}{2}(1, \sqrt{2}, 1)^T.$$

KEY IDEA

Sanity check — orthogonality is automatic. $q_1 \cdot q_2 = \frac{1}{2\sqrt{2}}(1 - 0 - 1) = 0$ and $q_1 \cdot q_3 = \frac{1}{4}(1 - 2 + 1) = 0$. The eigenvectors came out perpendicular *without us asking* — the Spectral Theorem working for free.

Step 3 — Assemble $A = Q\Lambda Q^T$.

$$Q = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 2 + \sqrt{2} & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 - \sqrt{2} \end{pmatrix}.$$

Because Q is orthogonal, $Q^{-1} = Q^T$ — you never invert anything.

Step 4 — The same answer as a sum of projections.

$$A = (2 + \sqrt{2}) q_1 q_1^T + 2 q_2 q_2^T + (2 - \sqrt{2}) q_3 q_3^T.$$

The dominant layer ($\lambda_1 \approx 3.414$) does most of the work; the $\lambda_3 \approx 0.586$ layer is a gentle finishing touch.

Problem. Spectrally decompose

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and write it as a projection sum.

Step 1 — Eigenvalues.

$$\det \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = (2-\lambda)^2 - 1 = 0 \Rightarrow \lambda = 3, 1.$$

Step 2 — Eigenvectors (perpendicular).

$$\lambda_1 = 3: x = y \Rightarrow q_1 = \frac{1}{\sqrt{2}}(1, 1)^T; \quad \lambda_2 = 1: x = -y \Rightarrow q_2 = \frac{1}{\sqrt{2}}(1, -1)^T.$$

Step 3 — Decomposition and projection sum.

$$A = 3 \underbrace{\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{q_1 q_1^T} + 1 \underbrace{\frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}}_{q_2 q_2^T} = \begin{pmatrix} 1.5 & 1.5 \\ 1.5 & 1.5 \end{pmatrix} + \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \checkmark$$

This is the matrix in the circle→ellipse figure above: stretch $\times 3$ along the 45° line, leave the 135° line untouched.

WATCH OUT

The “perpendicular eigenvectors, clean transpose” guarantee is *special to symmetric matrices*. The moment $A \neq A^T$, eigenvectors can tilt away from 90° and the tidy Q^T must be replaced by a genuine inverse P^{-1} . That is precisely the next rung of the ladder.

WHY IT MATTERS

Every **covariance matrix** is symmetric, so PCA *is* spectral decomposition: the top eigenvector is the direction of maximum variance. Quadratic forms $x^T A x$ (optimization, physics energy, statistics) become trivial in eigen-coordinates because the cross-terms vanish.

Part 2 | Eigen Decomposition

ANY SQUARE MATRIX — PERPENDICULARITY GONE, THE IDEA SURVIVES

Now we leave the perfect symmetric world. A general square matrix still has eigenvectors — stubborn directions it only scales — but they are **no longer perpendicular**. The “stretch along special axes” story is intact; we just lose the orthogonal luxury and pay with an honest matrix inverse.

INTUITION

Speak the matrix’s native language. A general matrix looks chaotic in the standard x, y grid because it both rotates *and* stretches. But it has its own private coordinate system — its eigenvectors — in which it does nothing but scale. The trick $A = P D P^{-1}$ is a **translator**: P^{-1} translates your vector *into* the matrix’s native eigen-language, D does the embarrassingly simple scaling there, and P translates back.

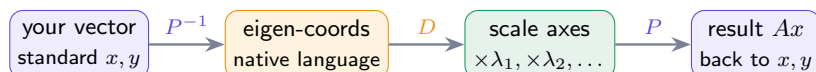
The decomposition: $A = P D P^{-1}$

Collect the (non-orthogonal) eigenvectors as columns of P and the eigenvalues on D . From $A P = P D$ with P invertible:

$$\boxed{A = P D P^{-1}} \quad P = [p_1 \ p_2 \ \cdots \ p_n], \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

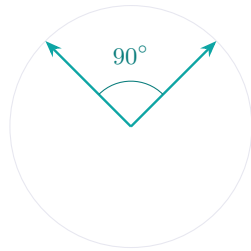
KEY IDEA

The crucial difference from Part 1. Spectral gave $A = Q \Lambda Q^T$ — the *transpose*, because orthonormal columns satisfy $Q^{-1} = Q^T$. Here the columns of P are **not** orthonormal, so $P^{-1} \neq P^T$ and we must compute a real inverse. *Spectral decomposition is just the lucky case of eigen decomposition where P happens to be orthogonal.* Same machine, one restriction removed.

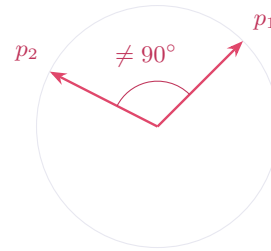


Eigen decomposition = a change of language. The hard transformation A becomes the trivial diagonal D once you speak the matrix’s own eigen-coordinates. Read $A = P D P^{-1}$ right-to-left.

Symmetric (Part 1)

eigenvectors $\perp \rightarrow$ clean Q^T

General square (Part 2)

eigenvectors tilt $\rightarrow$ need real P^{-1}

The single visual separating Part 1 from Part 2. Symmetric matrices keep their eigen-axes perpendicular; general matrices let them lean — which is exactly why the clean transpose becomes an inverse.

Problem.

Find $A = PDP^{-1}$ for $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ (note $A \neq A^T$).

Step 1 — Eigenvalues.

$\det \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix} = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 5, \lambda_2 = -1$. A **negative** eigenvalue is allowed (“flip + scale”) — something a positive-definite symmetric matrix never shows.

Step 2 — Eigenvectors.

$\lambda_1 = 5 : x = y \Rightarrow p_1 = (1, 1)^T$; $\lambda_2 = -1 : x = -2y \Rightarrow p_2 = (-2, 1)^T$. Angle check: $p_1 \cdot p_2 = -1 \neq 0$ — **not perpendicular**, as the figure warned.

Step 3 — Build P , D , and the genuine inverse.

$$P = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}, \det P = 3, P^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}, D = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}.$$

Verify $PDP^{-1} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = A$. ✓

Step 4 — The payoff: matrix powers become trivial.

Computing A^3 directly is tedious; with the decomposition the middle collapses since $P^{-1}P = I$:

$$A^3 = PD^3P^{-1}, D^3 = \begin{pmatrix} 125 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow A^3 = \begin{pmatrix} 41 & 84 \\ 42 & 83 \end{pmatrix}.$$

Cubing a diagonal matrix is just cubing three numbers — the trick behind matrix exponentials, Markov steady states, and systems of ODEs.

Problem. Diagonalize

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

Step 1 — Eigenvalues (shortcut).

For any triangular matrix the eigenvalues *are* the diagonal entries: $\det(A - \lambda I) = (2 - \lambda)(3 - \lambda) \Rightarrow \lambda_1 = 2, \lambda_2 = 3$ — no quadratic needed.

Step 2 — Eigenvectors.

$\lambda_1 = 2 : y = 0 \Rightarrow p_1 = (1, 0)^T$; $\lambda_2 = 3 : x = y \Rightarrow p_2 = (1, 1)^T$.

Step 3 — Assemble.

$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $A = PDP^{-1}$. Again $p_1 \cdot p_2 = 1 \neq 0$: a true eigen (not spectral) decomposition.

WATCH OUT

When eigen decomposition refuses to exist. (1) **Defective** matrices like $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ have a repeated eigenvalue but too few independent eigenvectors — P cannot be formed. (2) **Rotation** matrices like $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ rotate every direction, so no real vector stays on its line; eigenvalues are complex. Both failures vanish in Part 3: **SVD always exists**, for every matrix.

Part 3 | Singular Value Decomposition

EVERY MATRIX — SQUARE OR RECTANGULAR, ANY RANK — FACTORS INTO ROTATE · STRETCH · ROTATE

The boldest question in the lecture. Eigen decomposition needed a *square* matrix so input and output lived in the same space. But a 3×2 matrix eats 2-D vectors and outputs 3-D ones — “eigenvector” is meaningless, since Av cannot even be parallel to v . SVD’s answer is one of the most elegant ideas in all of linear algebra.

INTUITION

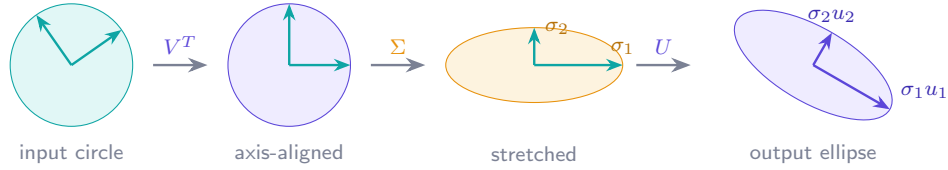
The sphere always becomes an ellipsoid. Feed any matrix the unit sphere of inputs; the output is always a perfect **ellipsoid**. SVD finds the perpendicular input directions (v_i) mapped to the perpendicular axes of that ellipsoid (u_i), and the axis lengths are the **singular values** σ_i . Eigen asked “which directions don’t rotate?” — too much for a rectangular matrix. SVD asks the always-answerable question: “which *perpendicular input frame* lands on a *perpendicular output frame*?”

The cleverness: use two orthonormal frames, not one

Eigen reused a single set of axes for input and output (hence “square”). SVD uses **two separate orthonormal frames** — input V and output U — with a pure diagonal stretch Σ bridging them:

$$\boxed{A = U \Sigma V^T} \quad \begin{array}{l} V \ (n \times n, \text{ orthonormal}) : \text{input axes (right-singular vectors)} \\ \Sigma \ (m \times n, \text{ diagonal } \geq 0) : \text{stretch factors } \sigma_i \\ U \ (m \times m, \text{ orthonormal}) : \text{output axes (left-singular vectors)} \end{array}$$

Right-to-left: V^T **rotates** the input so special directions align with the axes, Σ **stretches** each axis by σ_i (and pads dimensions with zeros), U **rotates** into the final pose.



The definitive SVD picture. A circle is rotated by V^T until special directions sit on the axes, stretched by Σ into an ellipse, then rotated by U into its final orientation. The singular values $\sigma_1 \geq \sigma_2$ are the ellipse's semi-axis lengths.

How to find U, Σ, V — the symmetric-matrix bridge

We reduce SVD to Part 1. Although A may be rectangular, both $A^T A$ and AA^T are **square and symmetric**:

$$A^T A = V \Sigma^T \Sigma V^T \Rightarrow \boxed{V = \text{eigenvectors of } A^T A, \sigma_i^2 = \text{its eigenvalues}}$$

$$AA^T = U \Sigma \Sigma^T U^T \Rightarrow \boxed{U = \text{eigenvectors of } AA^T}$$

And once you have V and the σ_i , each output axis is the normalized image of an input axis (from $AV = U\Sigma$):

$$\boxed{u_i = \frac{1}{\sigma_i} A v_i}.$$

KEY IDEA

Recipe. (1) Spectrally decompose the *smaller* of $A^T A$ or AA^T . (2) $\sigma_i = \sqrt{\lambda_i}$ (drop zeros). (3) Its eigenvectors give V (or U). (4) Get the other family free via $u_i = \frac{1}{\sigma_i} A v_i$. (5) Assemble $A = U \Sigma V^T$, padding Σ with zeros to match A 's shape.

Problem. Find the

SVD of $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$. *Smart move:* B is 3×2 , so $B^T B$ is only 2×2 — decompose the smaller one.

Step 1 — Build $B^T B$ and notice an old friend.

$$B^T B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

This is *exactly* the matrix from Part 1! Eigenvalues 3, 1; eigenvectors $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$.

Step 2 — Singular values and V .

$$\sigma_1 = \sqrt{3}, \sigma_2 = 1; \quad v_1 = \frac{1}{\sqrt{2}}(1, 1)^T, \quad v_2 = \frac{1}{\sqrt{2}}(1, -1)^T.$$

Step 3 — Get U for free via $u_i = \frac{1}{\sigma_i} B v_i$.

$$u_1 = \frac{1}{\sqrt{3}} B v_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad u_2 = \frac{1}{1} B v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

These are automatically orthonormal ($u_1 \cdot u_2 = 0$). A third column $u_3 = \frac{1}{\sqrt{3}}(1, -1, -1)^T$ completes U but rides on $\sigma_3 = 0$.

Step 4 — Assemble $B = U \Sigma V^T$.

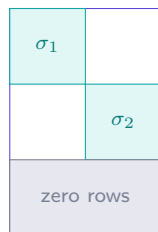
$$U = \begin{pmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad V^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Σ is 3×2 — the same shape as B — with a **zero row of padding**, because B maps 2-D into 3-D and the third output dimension gets nothing.

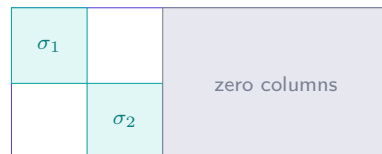
Why Σ has that odd zero-padding

Σ is always the same shape as A . It carries a diagonal block of singular values then pads with zeros: tall matrices pad with zero *rows* (output dimensions they cannot reach); wide matrices pad with zero *columns* (input directions they annihilate).

Tall A ($m > n$): Σ is $m \times n$



Wide A ($m < n$): Σ is $m \times n$



Σ inherits the shape of A : a diagonal block of singular values plus zero padding.

Problem. Find the SVD of $A =$

$$\begin{pmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{pmatrix}.$$

Step 1 — $A^T A$ and its eigenvalues.

$$A^T A = \begin{pmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \lambda = 6, 1, 0.$$

The single zero eigenvalue is expected: a 2×3 matrix has rank ≤ 2 , so one input direction is crushed.

Step 2 — Singular values and V .

$$\sigma_1 = \sqrt{6}, \sigma_2 = 1; \quad v_1 = \frac{1}{\sqrt{30}} \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad v_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}.$$

(v_3 is the $\lambda = 0$ eigenvector — the direction A annihilates.)

Step 3 — Left vectors via $u_i = \frac{1}{\sigma_i} A v_i$.

$$u_1 = \frac{1}{\sqrt{5}}(1, -2)^T, \quad u_2 = \frac{1}{\sqrt{5}}(2, 1)^T.$$

Step 4 — Assemble $A = U \Sigma V^T$.

$$U = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{6} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad V^T = \begin{pmatrix} 5/\sqrt{30} & -2/\sqrt{30} & 1/\sqrt{30} \\ 0 & 1/\sqrt{5} & 2/\sqrt{5} \\ -1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix}.$$

This time Σ is 2×3 with a **zero column** — the wide-matrix padding.

Problem. Find the SVD of

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \text{ (square, but } A \neq A^T \text{).}$$

Step 1 — $A^T A$ and eigenvalues.

$$A^T A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix} \Rightarrow \lambda = 45, 5.$$

Step 2 — Singular values and V .

$$\sigma_1 = \sqrt{45} = 3\sqrt{5}, \sigma_2 = \sqrt{5}; \quad v_1 = \frac{1}{\sqrt{2}}(1, 1)^T, \quad v_2 = \frac{1}{\sqrt{2}}(1, -1)^T.$$

Step 3 — U via $u_i = \frac{1}{\sigma_i} A v_i$.

$$u_1 = \frac{1}{\sqrt{10}}(1, 3)^T, \quad u_2 = \frac{1}{\sqrt{10}}(3, -1)^T.$$

KEY IDEA

Myth-buster. The eigenvalues of this A are 3 and 5, but its singular values are $3\sqrt{5} \approx 6.7$ and $\sqrt{5} \approx 2.2$. **For non-symmetric matrices, singular values $\neq$ eigenvalues.** They coincide only when A is symmetric positive-definite — the Part 1 case.

SVD vs. Eigen — the family resemblance

	Eigen / Spectral	SVD
Form	$A = PDP^{-1}$ (or $Q\Lambda Q^T$ if symmetric)	$A = U\Sigma V^T$
Works on	Square matrices with enough eigenvectors	Every matrix, any shape, always
Frames	One set of axes (in = out)	Two sets: input V , output U
Orthogonality	P generally not orthogonal	U, V always orthonormal
Diagonal entries	Eigenvalues — can be $\pm$ or complex	Singular values — always real ≥ 0
Outer factors	P and P^{-1} are inverses	U and V generally unrelated

The link: U = eigenvectors of AA^T , V = eigenvectors of A^TA , $\sigma_i = \sqrt{\lambda_i(A^TA)}$.

Part 4 | Low-Rank Approximation & the Spectral Norm

THE PAYOFF — WHY SVD IS THE MATHEMATICS BEHIND COMPRESSION

Remember how a symmetric matrix split into importance-ranked projection layers? SVD does the same for *every* matrix — and this is where it earns its place as arguably the most useful factorization in applied mathematics.

A matrix is a stack of rank-1 layers

Multiplying $U\Sigma V^T$ out, Σ pairs each u_i with its v_i and scales by σ_i . The matrix becomes a sum of r (= rank) **rank-1 layers**, sorted loudest first:

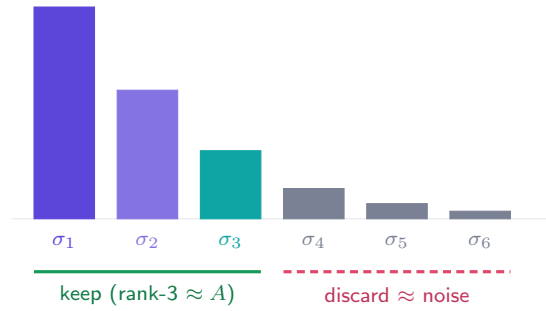
$$A = \sum_{i=1}^r \sigma_i u_i v_i^T = \underbrace{\sigma_1 u_1 v_1^T}_{\text{loudest}} + \underbrace{\sigma_2 u_2 v_2^T}_{\text{quieter}} + \cdots + \underbrace{\sigma_r u_r v_r^T}_{\text{faintest}}.$$

Because $\sigma_1 \geq \sigma_2 \geq \cdots$, the first few layers carry almost all the energy. **Truncate at k terms** for the best rank- k summary: $\hat{A}(k) = \sum_{i=1}^k \sigma_i u_i v_i^T$.

KEY IDEA

Eckart–Young theorem. Of *all* rank- k matrices, this truncated SVD is provably the closest to A (in both spectral and Frobenius norms). You cannot do better with rank k , and the error is dictated exactly by the singular values you discarded.

Singular-value “energy” — keep the tall bars, discard the stubs



The compression idea in one glance. Most of a real matrix’s content lives in its first few singular values; the long tail is faint detail or noise. Keeping the loud layers and dropping the quiet ones is exactly how SVD-based image compression, denoising, and dimensionality reduction work.

Problem. Compress $B =$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ to rank 1 using its SVD (from Part 3).}$$

Step 1 — Take only the loudest layer.

Keep $\sigma_1 = \sqrt{3}$ with $u_1 = \frac{1}{\sqrt{6}}(2, 1, 1)^T$ and $v_1^T = \frac{1}{\sqrt{2}}(1, 1)$.

Step 2 — Form $\hat{B}_1 = \sigma_1 u_1 v_1^T$.

$$\hat{B}_1 = \sqrt{3} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} (1 \quad 1) = \frac{1}{2} \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}.$$

Step 3 — How good is it? The error is the discarded singular value.

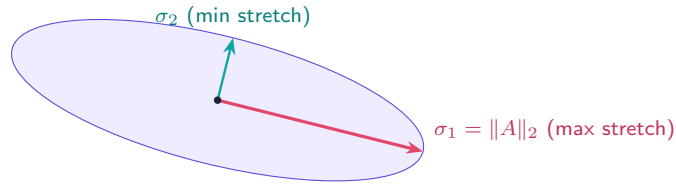
We dropped $\sigma_2 = 1$, so $\|B - \hat{B}_1\|_2 = \sigma_2 = 1$ and $\|B - \hat{B}_1\|_F = \sqrt{\sigma_2^2} = 1$. The rank-1 layer already captures the dominant structure of B .

The spectral norm: a matrix’s maximum stretch

How “big” is a matrix — the most it can ever amplify a vector? That single number is the **spectral norm**, and SVD answers it instantly:

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1.$$

It is just the longest axis of the output ellipsoid. The input achieving it is v_1 ; the output points along u_1 .



The spectral norm is the ellipsoid's longest semi-axis σ_1 — the worst-case amplification, central to stability, conditioning, and error bounds.

Problem. Find $\|A\|_2$ for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

The spectral norm is the top singular value. From $A^T A = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$, eigenvalues solve $\lambda^2 - 30\lambda + 4 = 0 \Rightarrow \lambda = 15 \pm \sqrt{221}$. Then

$$\sigma_1 = \sqrt{15 + \sqrt{221}} \approx \sqrt{29.87} \approx 5.465, \quad \|A\|_2 \approx 5.465.$$

(The smaller singular value is $\sigma_2 \approx 0.366$, so this matrix stretches its best direction $\sim 15\times$ more than its worst — fairly ill-conditioned.)

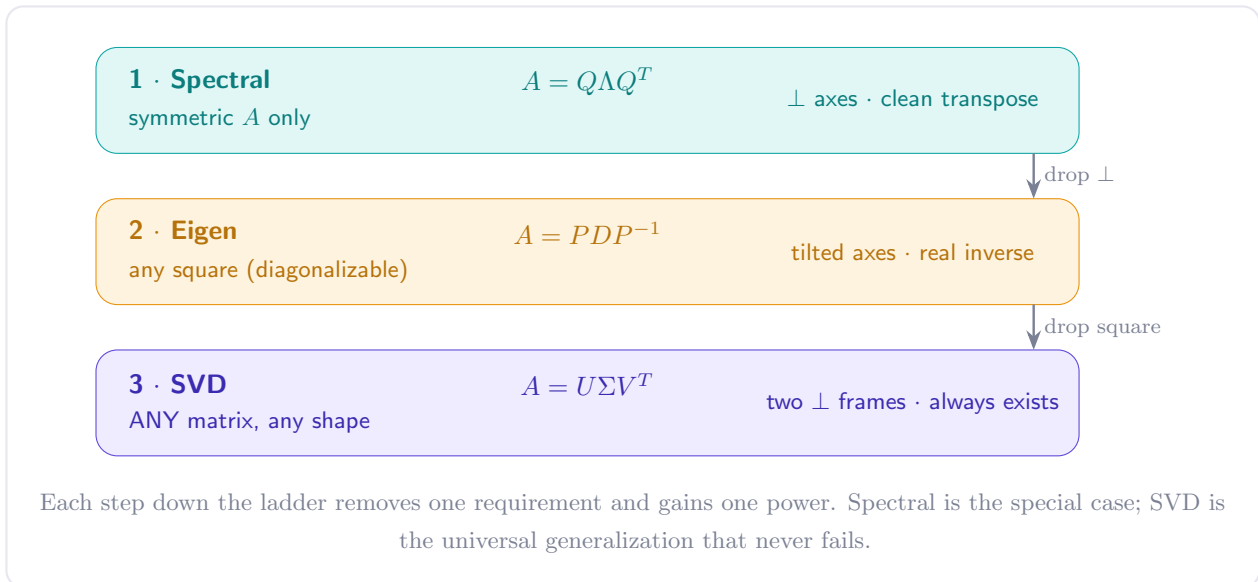
WHY IT MATTERS

A 1000×1000 image is a million numbers. If its singular values decay fast (they usually do), keeping the top 50 layers reconstructs it almost perfectly using only $\sim 10\%$ of the storage. The same truncation removes noise (noise hides in tiny singular values), drives **PCA** dimensionality reduction, and builds the latent factors in **recommender systems**.

Part 5 | The Unifying Picture

ONE PAGE TO RULE THEM ALL

All three decompositions answer one question — “*what simple moves is this matrix made of?*” — under progressively weaker assumptions.



Which one should I reach for?

If your matrix is...	Use...	Because
Symmetric (covariance, Gram, Laplacian)	Spectral $Q\Lambda Q^T$	Orthonormal eigenvectors, real eigenvalues, the cleanest decomposition — and it is PCA.
Square; you need powers / e^A / dynamics	Eigen PDP^{-1}	$A^k = PD^kP^{-1}$ turns repeated multiplication into scalar powers.
Rectangular, low-rank, or you want compression	SVD $U\Sigma V^T$	Always exists; gives the best rank- k approximation and the spectral norm.
Non-diagonalizable / complex eigenvalues	SVD	Real, non-negative σ 's sidestep eigen decomposition's failure modes.

KEY IDEA

The thirty-second recap. A matrix is a transformation. Decomposition rewrites it as **rotate** → **stretch** → **rotate**. When the matrix is symmetric, the two rotations are transposes of each other and the axes are perpendicular (**spectral**). For a general square matrix the axes tilt, so we swap the transpose for an inverse (**eigen**). For any matrix at all, we use two independent perpendicular frames joined by a non-negative stretch (**SVD**) — and sorting that stretch by size hands us compression, denoising, and the spectral norm for free.

Companion to ZC416 — Mathematical Foundations for ML / Data Science · Lecture 5: Matrix Decompositions. Built as an intuition-first study guide; all worked examples verified numerically.